

Airbnb Dynamic Pricing Analysis

Python | Excel | Tableau

1. Project Overview

This project analyzes Airbnb listing data to understand how factors such as **room type, property size, guest capacity, number of bedrooms, property type, and location** are associated with nightly prices.

The objective was to identify meaningful pricing patterns that could help hosts better understand the market and make more informed pricing decisions.

2. Dataset Overview

The final dataset contains **3,382 Airbnb listings across 48 locations**.

Key fields used in the analysis included:

- Basic Night Price
- Room Type
- Property Type
- Property Size
- Person Capacity
- Bedrooms
- City / Location
- Star Rating
- Airbnb Listing ID

The overall **median nightly price was ₹2,814**, while the **average star rating was 4.68**.

3. Data Preparation & Analysis

The dataset was cleaned and prepared before visualization. This included checking missing values, reviewing inconsistent categories, creating cleaner room/property classifications, examining price distributions and outliers, and validating important numerical fields.

Median nightly price was used for most pricing comparisons because Airbnb pricing can contain expensive outliers that may distort the average.

The cleaned data was then analyzed and visualized using **Tableau**, with additional exploratory analysis performed using **Python**.

4. Key Findings

Room Type: Entire-home listings recorded a median nightly price of approximately **₹3,110**, compared with **₹2,104 for private rooms** and **₹1,110 for shared rooms**. This suggests guests generally pay a premium for greater privacy and exclusive use of a property.

Guest Capacity: Nightly prices generally increased as guest capacity increased. For example, median prices rose from approximately **₹1,419 for one guest** to substantially higher levels for larger-capacity properties. Some higher-capacity groups had fewer listings, so these values should be interpreted carefully.

Property Size: Small and medium properties represented most of the market, with **1,441 small** and **1,428 medium** listings, compared with **586 large properties**.

Bedrooms: Median nightly prices generally increased as the number of bedrooms increased, although the relationship was not perfectly linear. The final analysis focused on **1–6 bedroom properties** to reduce the influence of very small sample groups.

Location: Among the top locations selected by listing volume, median prices varied considerably. For example, **Donaje (₹7,639)** and **Pirangut (₹7,016)** had higher median prices than **Pune (₹2,820)**. These results should not be interpreted as Donaje or Pirangut having more listings than Pune; the locations were selected based on listing volume and then compared by median price.

Property Type: Common property types such as entire homes and apartments were analyzed separately from rare premium property types, helping prevent unusually expensive but low-volume categories from dominating the conclusions.

5. Business Insights & Recommendations

The analysis indicates that Airbnb pricing is associated with a combination of **capacity, privacy, property characteristics, and location**, rather than any single factor.

Hosts could use comparable listings with similar room type, capacity, bedroom count, property type, and location as a starting point when setting prices. Larger properties and entire-home listings may support higher prices, but pricing decisions should also consider local competition and listing volume.

Premium prices observed in categories with few listings should be treated cautiously because a small number of properties can produce unstable pricing patterns.

6. Dashboard

The final Tableau dashboard includes four headline KPIs:

Median Nightly Price: ₹2,814 | Total Listings: 3,382 | Average Rating: 4.68 | Total Locations: 48

It also provides analysis of pricing by **room type, guest capacity, bedrooms, property type and location**, alongside listing distribution by property size.

7. Conclusion

This project demonstrates how Airbnb listing data can be transformed into practical pricing insights through **data cleaning, exploratory data analysis, validation, and interactive visualization**.

The analysis suggests that larger capacity, greater privacy, property characteristics, and location are important factors associated with nightly prices. At the same time, listing volume and sample size should always be considered before drawing pricing conclusions.